Parametric Evaluation of Glazing Systems and Their Influence on Building Cooling Energy and Solar Heat Gain

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ABSTRACT

Glazing selection affects transmitted solar radiation, conductive heat transfer, indoor thermal conditions, and mechanical cooling demand. This study evaluates four external-window configurations—single clear, double clear, double low-emissivity (Low-E), and triple clear glazing—using a controlled DesignBuilder/EnergyPlus simulation in which building geometry, envelope construction, schedules, HVAC configuration, weather data, and a 30% window-to-wall ratio were held constant. The model used a packaged-DX cooling system with a seasonal coefficient of performance of 2.5 and detailed residential occupancy, lighting, infiltration, ventilation, and domestic-hot-water settings. Annual and monthly cooling electricity, total cooling-load magnitude, window solar heat gain, and operative temperature were compared. Relative to single clear glazing, double clear reduced annual solar gain by 26.79% but slightly increased cooling electricity by 0.04%. Double Low-E yielded the best combined performance, reducing annual solar gain by 35.02% and annual cooling electricity by 3.16%, while triple clear achieved reductions of 30.05% and 1.04%, respectively. The results demonstrate that strong solar-gain suppression does not necessarily produce a proportional reduction in cooling electricity and that glazing should be assessed as part of the complete building-energy system.

Keywords: Glazing, Low-E Glass, DesignBuilder, EnergyPlus, Cooling Energy

1. Introduction

Buildings represent a major end-use sector for energy, and a considerable portion of their operational demand is associated with maintaining acceptable indoor thermal conditions [1,2]. For this reason, the thermal design of the building envelope is central to efforts aimed at reducing cooling and heating requirements. Among envelope components, windows are particularly important because they are simultaneously pathways for conductive heat transfer and for the transmission of solar radiation. Unlike an opaque wall, a glazed opening can therefore influence the building heat balance through several coupled mechanisms, including solar transmission, indoor–outdoor temperature difference, long-wave radiative exchange, and interaction with the HVAC system. In cooling-dominated operation, these mechanisms can strongly affect peak loads, annual electricity use, and the indoor operative temperature.

Glazing performance is governed not only by the number of panes but also by the thermal and optical characteristics of the complete window system. Increasing the number of panes generally increases thermal resistance by introducing additional glazing layers and sealed air spaces, while low-emissivity (Low-E) coatings reduce radiative heat exchange and modify the amount of solar energy transmitted through the window. The resulting energy response is not necessarily straightforward: a glazing system that substantially suppresses window solar heat gain may not produce an equivalent percentage reduction in whole-building cooling electricity. The final HVAC demand is determined by the combined contribution of window heat transfer, opaque-envelope conduction, infiltration, ventilation, lighting, occupancy, equipment gains, schedules, control setpoints, and system efficiency. Thus, solar-control performance should be interpreted within the complete building-energy balance rather than as an isolated indicator.

A broad range of previous studies has demonstrated that fenestration design can materially affect building performance. Important parameters include window-to-wall ratio (WWR), orientation, glazing construction, and the use of external shading or solar-control measures [3–7]. Investigations of WWR in different climatic contexts have shown that the preferred window area depends strongly on the interaction between solar exposure, heat transfer, and local operating conditions [3,5–7]. Similarly, studies of fenestration geometry and shading have reported meaningful changes in energy consumption when the solar exposure of windows is altered [4]. These findings establish the importance of window design, but they also indicate that glazing options should be compared under controlled conditions so that the influence of the glazing itself is not confounded by simultaneous changes in geometry, occupancy, HVAC operation, or other envelope parameters.

Building-energy simulation provides a practical framework for such controlled comparisons because the same geometry and operating schedules can be retained while individual construction parameters are varied. EnergyPlus is a widely used whole-building simulation engine capable of resolving the interacting heat-balance and HVAC processes required for this type of analysis [8]. In the present work, DesignBuilder was used as the modeling interface with EnergyPlus as the simulation engine. The comparison was deliberately structured as a parametric study in which the building geometry, envelope construction, activity template, occupancy and lighting schedules, infiltration, ventilation, HVAC configuration, weather data, and window layout were kept unchanged. A window-to-wall ratio of 30% was maintained in every case, and only the external glazing template was modified.

Accordingly, the present study evaluates four practically distinct glazing configurations: single clear, double clear, double Low-E, and triple clear glazing. Their performance is compared using annual and monthly cooling electricity, total cooling-load magnitude, solar heat gain through exterior windows, and operative temperature. Particular attention is given to the relationship between solar-gain reduction and realized cooling-energy saving, because these two indicators need not vary proportionally. By holding the non-glazing

inputs constant, the analysis aims to isolate the system-level effect of glazing choice and to identify which configuration provides the most favorable balance between solar control and cooling-energy performance under the simulated residential operating conditions.

2. Experimental Setup/Numerical Modeling

2.1 Building-Energy Model

The building was modeled in DesignBuilder with EnergyPlus as the simulation engine [8]. Figure 1 presents the model used for the parametric comparison. The same geometry, constructions, activity template, schedules, lighting, HVAC system, ventilation settings, weather file, and window layout were used in all cases; only the external glazing template was changed.

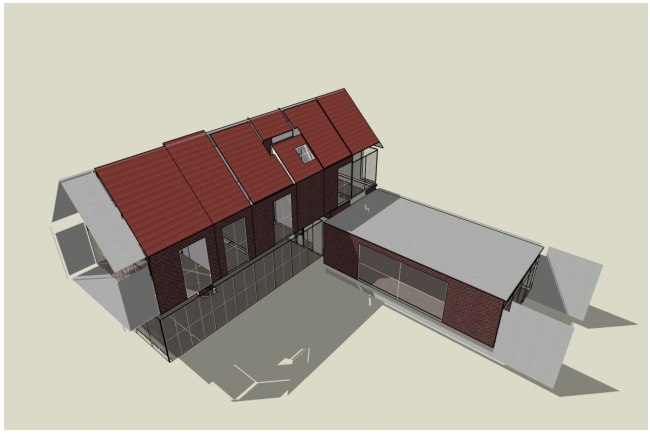


Figure 1 Building model used for the glazing simulations.

The envelope constructions implemented during model setup are listed in Table 1. The documented setup also records reference material properties for lime concrete of thermal conductivity 0.35 $W\,m^{-1}\,K^{-1}$, specific heat 900 $J\,kg^{-1}\,K^{-1}$, and density 1600 $kg\,m^{-3}$, and for the RCC slab of 1.58 $W\,m^{-1}\,K^{-1}$, 1000 $J\,kg^{-1}\,K^{-1}$, and 2500 $kg\,m^{-3}$, respectively.

Table 1 Envelope construction details and corresponding DesignBuilder layer visualizations used in the model.

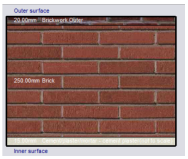
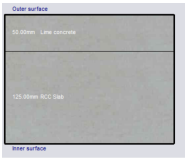
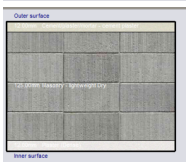
Construction	Layer/material	Thickness	Construction image
External wall	Brickwork Outer / Brick / Cement plaster-mortar	20 / 250 / 15 mm	
Flat roof	Lime concrete / RCC slab	50 / 125 mm	
Pitched occupied roof	Roof tile / XPS insulation board / RCC slab	20 / 60 / 125 mm	
Interior partition	Cement plaster / lightweight dry masonry / dense plaster	12 / 125 / 12 mm	

Table 1 (continued). Envelope construction details and corresponding DesignBuilder layer visualizations used in the model.

Construction	Layer/material	Thickness	Construction image
Ground floor	Sand and gravel / cast concrete / ceramic-clay floor tile	150 / 100 / 20 mm	

Table 2 Documented reference thermophysical properties available in the simulation setup notes.

Material	k (W/m K)	c_p (J/kg K)	ρ (kg/m^3)
Lime concrete	0.35	900	1600
RCC slab	1.58	1000	2500

2.2 Activity, Loads, Ventilation and HVAC Settings

Table 3 summarizes the principal operational inputs documented in the simulation setup. The activity template was *Domestic Lounge* with an occupancy density of 0.05 person/m^2. Separate residential occupancy schedules were configured: the weekday profile was 07:00–09:00 and 16:00–23:00, while the weekend/Friday–Saturday profile was 07:00–23:00. The same relevant dwelling schedules were linked to HVAC and lighting controls. Model infiltration was set to 0.5 air changes per hour and operated continuously. Natural ventilation was enabled in the project setup, while mechanical ventilation was also active in the packaged-DX template using the minimum-fresh-air method based on per-person and per-area requirements.

Table 3 Operational, internal-load, ventilation, HVAC and DHW settings retained for all glazing cases.

Category	Parameter	Model setting
Activity/occupancy	Template; density	Domestic Lounge; 0.05 person/m^2
Schedules	Occupancy	Weekday: 07:00–09:00 and 16:00–23:00; weekend/Friday–Saturday: 07:00–23:00
Infiltration	Constant rate	0.5 ac/h; On 24/7
Heating	Setpoint / setback	20 / 15 °C
Cooling	Setpoint / setback	24 / 28 °C
Lighting	Power density	3 W/m^2 per 100 lux; Dwell_DomLounge_Light
Lighting	Luminaire/fractions	2-Surface mount; return-air 0; radiant 0.72; visible 0.18; convective 0.10
Natural ventilation	Status	Enabled in the simulation setup
Mechanical ventilation	Status / outside air	On; 4-Min fresh air (sum per person + per area); Dwell_DomLounge_Occ
HVAC	Template/system	Packaged DX; default cooling system; electricity from grid
HVAC	Seasonal COP	2.500
Auxiliary energy	Pump energy	0.0000 W/m^2; Dwell_DomLounge_Occ
Cooling operation	Schedule	Dwell_DomLounge_Cool
DHW	Template/type	Project DHW; instantaneous hot water only
DHW	COP/fuel	0.8500; electricity from grid
DHW	Delivery/mains	65 / 22 °C

2.3 Glazing Parametric Cases

The window layout was fixed at 30% WWR using the DesignBuilder "preferred height" method. Window height was 1.50 m, spacing 5.00 m, sill height 0.80 m, and outside reveal depth 0 m. No shading device was activated for the glazing comparison. Table 4 lists the four glazing templates exactly as documented in the setup.

Table 4 Glazing cases used for the parametric simulation.

Case	DesignBuilder glazing definition
Single clear	Sgl Clr 6mm; clear, no shading
Double clear	Dbl Clr 6mm/13mm Air; no shading
Double Low-E	Dbl LoE (e2=.1) Clr 3mm/13mm Air
Triple clear	Trp Clr 3mm/6mm Air; no shading

The evaluated outputs were cooling electricity, total cooling-load magnitude, solar heat gain through exterior windows, heating energy, and operative temperature. Annual cooling-energy saving relative to single clear was calculated

as

$$S_E = \frac{E_{SC} - E_i}{E_{SC}} \times 100\%, \quad (1)$$

where E_{SC} is the annual cooling electricity of the single-clear baseline and E_i is that of glazing option i. Solar-gain reduction was determined analogously.

3. Results and Discussion

3.1 Annual Performance

Table 5 summarizes the annual outputs. Single clear required 40,683.24 kWh of cooling electricity. Double clear was effectively unchanged at 40,699.96 kWh, whereas double Low-E reduced cooling electricity to 39,396.43 kWh, a 3.16% saving. Triple clear reduced cooling electricity by 1.04%. Annual heating energy was approximately 1–2 kWh in all cases, showing that the simulated operating regime was strongly cooling dominated.

Table 5 Annual performance of the four glazing cases.

Glazing	Cooling elec. (kWh)	Saving (%)	Solar gain (kWh)	Solar reduction (%)	Cooling load (kWh)	Mean op. temp. (°C)
Single clear	40,683.24	0.00	125,295.25	0.00	101,708.08	30.42
Double clear	40,699.96	-0.04	91,722.95	26.79	101,749.88	30.89
Double Low-E	**39,396.43**	**3.16**	**81,416.54**	**35.02**	**98,491.08**	30.76
Triple clear	40,258.15	1.04	87,650.23	30.05	100,645.37	30.88

Figures 2 and 3 show the annual cooling-electricity and solar-gain results. The main finding is the non-proportional relationship between solar-gain reduction and realized cooling-energy saving. Double clear reduced annual solar gain by 26.79% but slightly increased cooling electricity. Triple clear reduced solar gain by 30.05% while yielding only a 1.04% cooling saving. Double Low-E was the only alternative providing the largest improvement in both indicators.

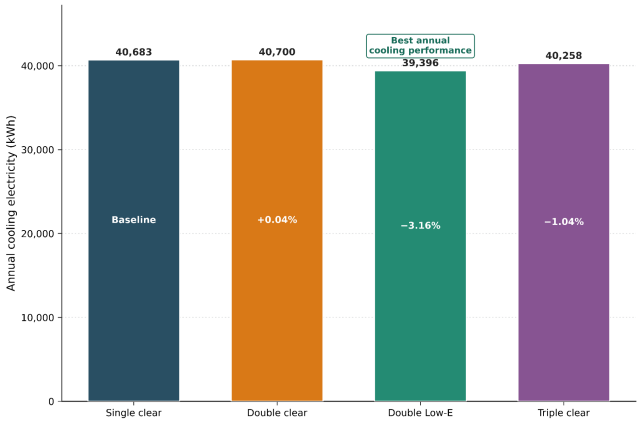


Figure 2 Annual cooling-electricity comparison.

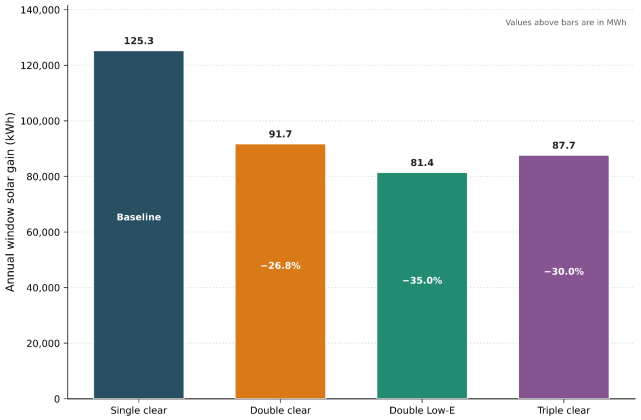


Figure 3 Annual window solar-gain comparison.

3.2 Seasonal Cooling Response

Monthly cooling electricity rose from the winter period to a maximum in May and remained high through the warm season before declining sharply in November and December (Fig. 4). The total cooling-load magnitude followed the same seasonal structure (Fig. 5). Double Low-E generally produced the lowest cooling demand during the cooling-intensive months. Because the same cooling-system COP was retained for all cases, the close similarity of cooling-load and cooling-electricity rankings is physically consistent.

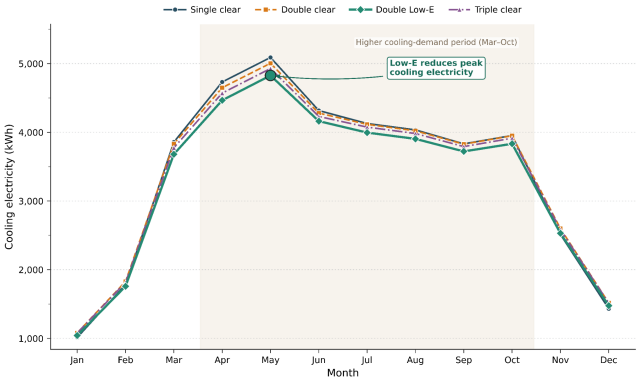


Figure 4 Monthly cooling electricity for the four glazing systems.

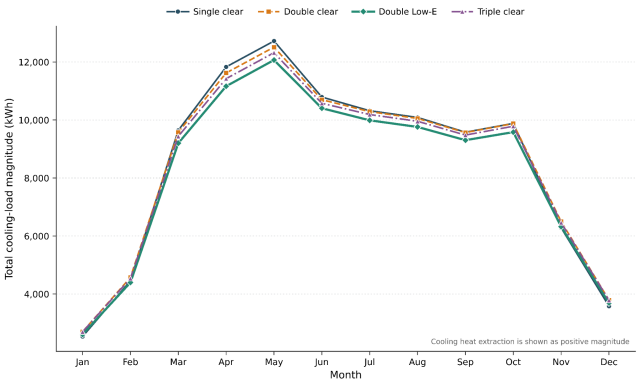


Figure 5 Monthly total cooling-load magnitude for the four glazing systems.

3.3 Solar Gain and Operative Temperature

Figure 6 shows that glazing choice produced a much larger separation in window solar heat gain than in cooling electricity. Relative to single clear, the annual solar-gain reductions were 26.79% for double clear, 35.02% for double Low-E, and 30.05% for triple clear. This large optical/solar effect is

not transferred one-to-one into HVAC electricity because the building cooling balance includes multiple non-window heat-flow and internal-load terms.

The monthly operative-temperature trends (Fig. 7) generally followed outdoor seasonal temperature. The annual mean operative temperatures were 30.42, 30.89, 30.76, and 30.88 °C for single clear, double clear, double Low-E, and triple clear, respectively. These annual means are reported as diagnostic simulation outputs and should not be interpreted as a complete thermal-comfort assessment, which would require occupied-hour distributions, humidity, air speed, and an appropriate comfort model.

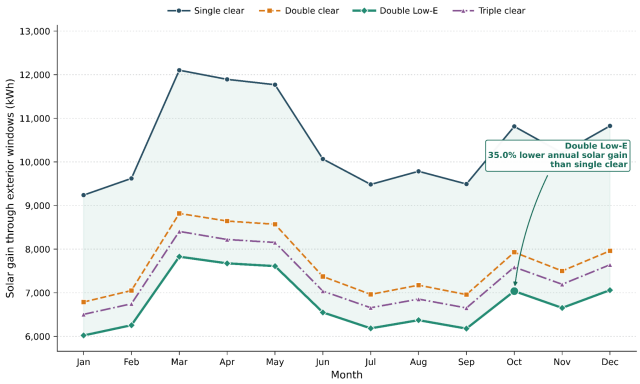


Figure 6 Monthly solar gain through exterior windows.

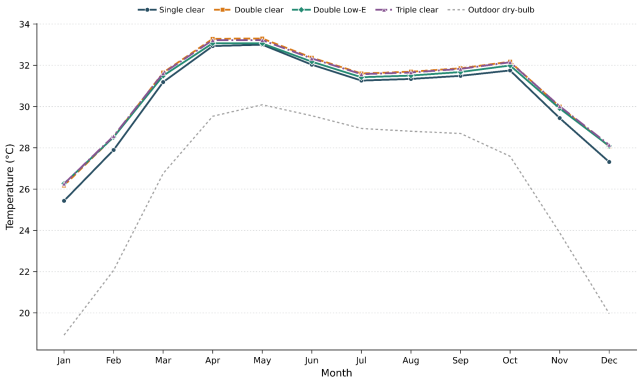


Figure 7 Monthly operative temperature and outdoor dry-bulb temperature.

3.4 Solar-Control Effectiveness Versus Cooling Saving

The relationship between solar-control effectiveness and cooling-energy benefit is summarized in Figs. 8 and 9. The double-clear case is particularly informative: a substantial 26.8% reduction in annual solar gain produced essentially zero cooling-electricity saving. By contrast, double Low-E combined the greatest solar-gain reduction (35.0%) with the greatest cooling-electricity reduction (3.16%). The normalized comparison likewise shows that differences in cooling electricity and cooling load are modest compared with the differences in solar gain. Thus, selecting glazing only from transmitted-solar-energy performance can be misleading for whole-building energy decisions.

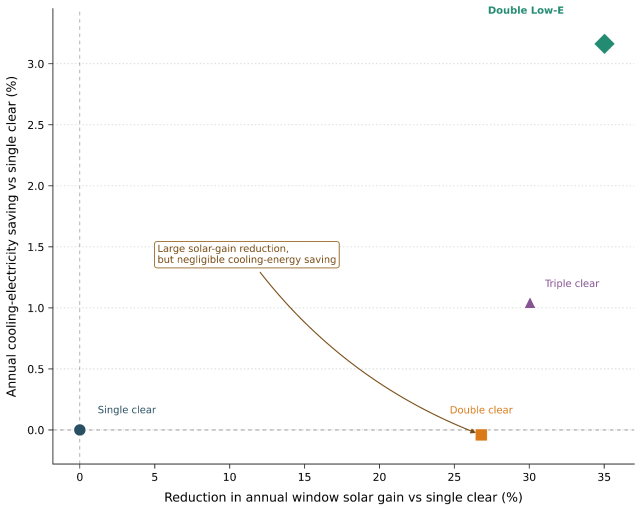


Figure 8 Solar-gain reduction versus annual cooling-electricity saving.

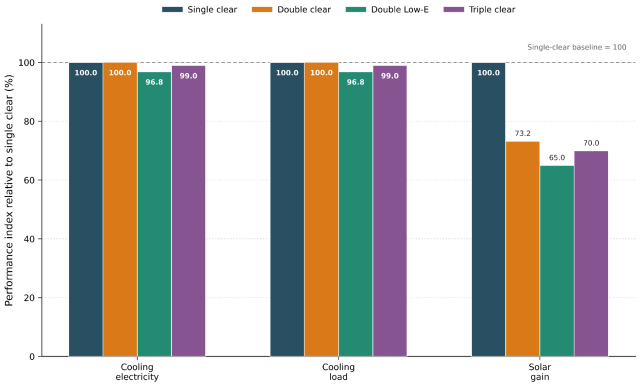


Figure 9 Normalized annual performance relative to single clear.

4. Conclusions

A controlled DesignBuilder/EnergyPlus study was performed to determine the effect of glazing type while keeping geometry, 30% WWR, envelope materials, residential schedules, infiltration, ventilation, lighting, HVAC configuration, and weather conditions unchanged. Among the four investigated alternatives, double Low-E provided the best overall performance, reducing annual cooling electricity by 3.16% and annual window solar gain by 35.02% relative to single clear. Triple clear produced a smaller 1.04% cooling saving despite a 30.05% reduction in solar gain, while double clear reduced solar gain by 26.79% but did not reduce annual cooling electricity.

The findings confirm that reductions in window solar heat gain do not translate proportionally into whole-building cooling-energy savings, highlighting the importance of evaluating glazing as part of the complete building-energy system rather than solely on the basis of solar-control performance. Under the simulated residential operating conditions, double Low-E glazing achieved the most favorable overall performance, combining the largest reduction in annual solar heat gain with the greatest decrease in cooling electricity. In contrast, the comparatively small cooling-energy benefits of double-clear and triple-clear glazing, despite their substantial reductions in solar gain, demonstrate that overall cooling demand is governed by the combined effects of solar transmission, conductive heat transfer, internal loads, ventilation, infiltration, and HVAC operation. Overall, among the glazing configurations examined, double Low-E glazing provided

the most effective balance between solar-gain control and cooling-energy reduction.

5. References

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